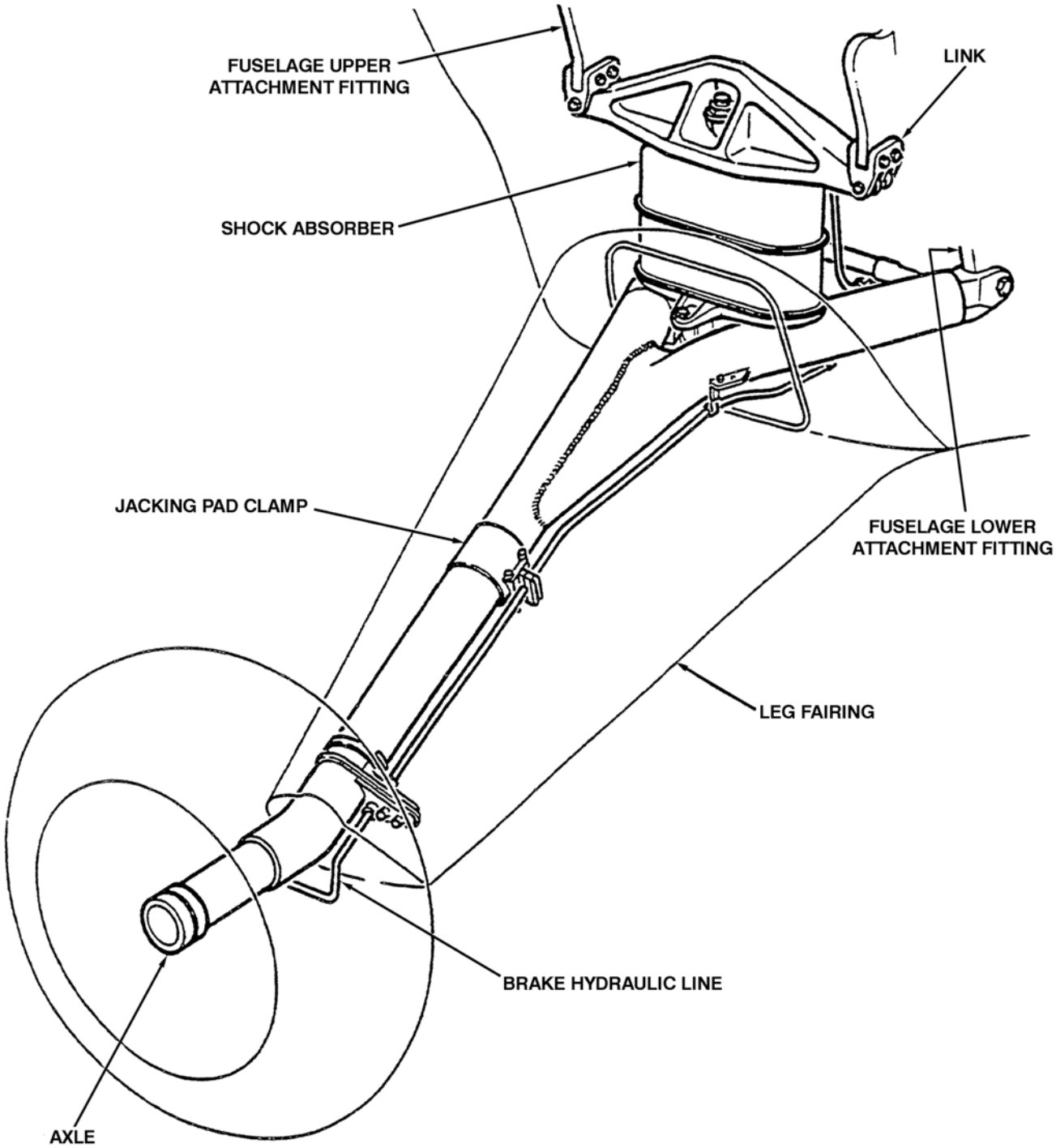
Main Gear – General Data

1. General

Each main gear consists of a leg assembly, shock absorber, and wheel and brake assembly. The leg assembly is bolted to two lower attachment fittings on the fuselage, and these provide the pivoted point for leg movement. Restriction and damping of leg movement is controlled by a shock absorber of the urethane block type. Each shock absorber is bolted to two fuselage upper attachment fittings and to metal pads on the leg assembly. Installed on the axle is a wheel and a hydraulically-operated brake assembly. A two piece fairing encloses the leg. Refer to Figure 1 for details.

A. Shock Absorber

The shock absorber consists essentially of urethane blocks, and absorbs the normal operating compression and rebound loads. The compression unit consists of an upper and lower platen between which are sandwiched two compression urethane blocks interspaced with a separator plate. Below the lower platen is the rebound unit comprising a rebound urethane block and end plate. The units are preloaded by a preload bolt and fork fitting. On landing, the leg is forced up against the compression blocks and upper platen which absorb the impact force, then as the leg rebounds, the rebound force is absorbed by the rebound block.



Main Gear
Figure 1

2. Removal/Installation

Refer to Figure 2 and Figure 3 for details.

A. Remove Shock Absorber

1) Remove main gear/fuselage fairings.

2) Jack aircraft until wheel is almost clear of ground (Refer to 07-00-00, Lifting and Shoring – General Data).

3) Remove cotter pin from preload bolt and remove preload nut.

4) Remove three bolts from underside of end plate and withdraw preload bolt.

5) Raise aircraft higher on jacks, and lift out compression blocks and separator plate.

B. Install Shock Absorber

1) Jack aircraft sufficiently, and position compression blocks and separator plate on lower platen.

2) Smear preload bolt, fork fitting and lower end of preload nut with grease, Specification MIL–G–21164. Assemble preload bolt and preload nut to main gear. Tighten three bolts on underside of end plate.

Note Bonding of urethane rebound block to metal end plates by operators is prohibited. A rebound block must be renewed, as necessary, with a complete assembly. Part No. C6U1137–3.

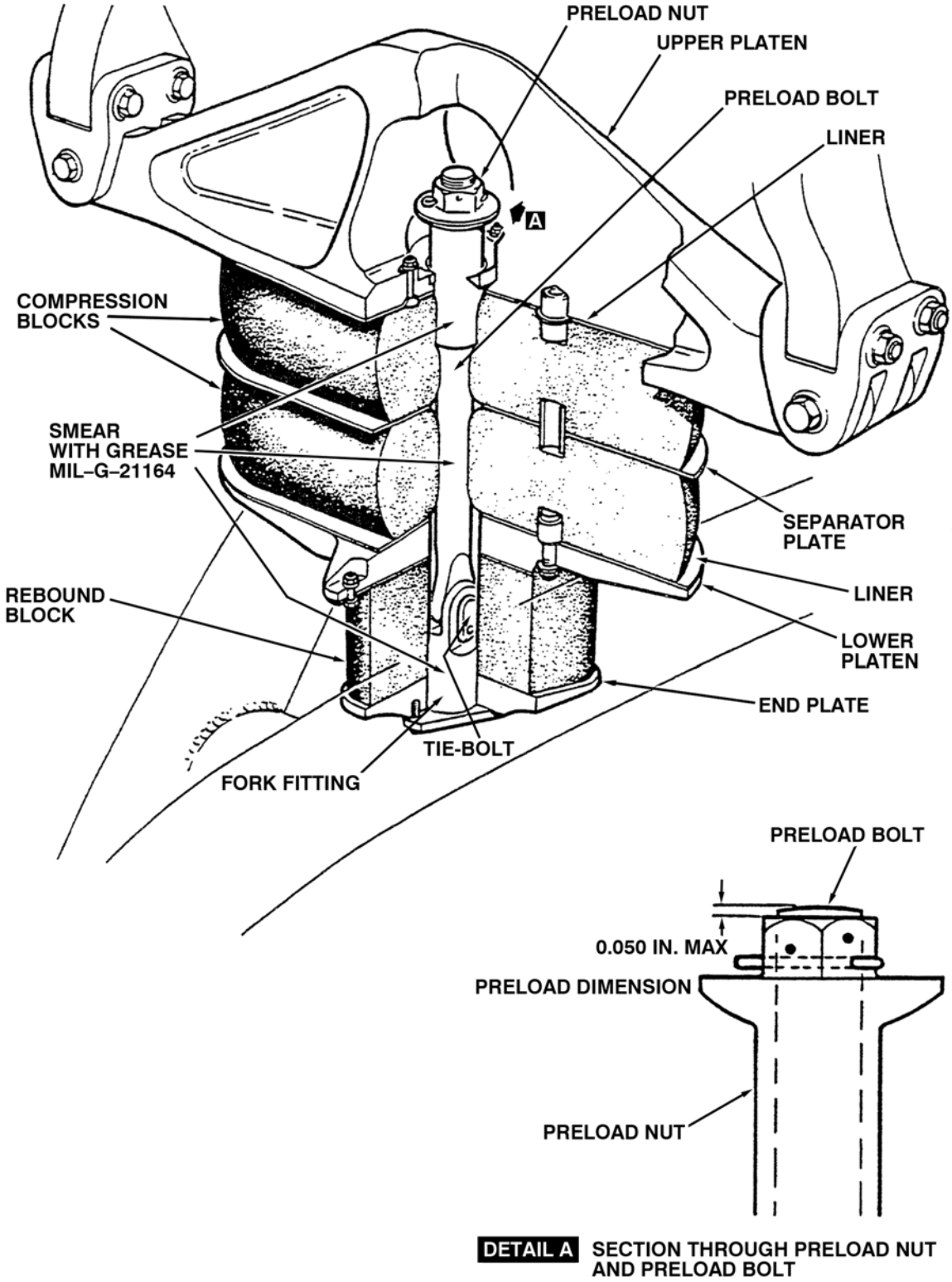
CAUTION - ENSURE THAT PRELOAD BOLT DOES NOT PROTRUDE MORE THAN 0.050 INCH THROUGH PRELOAD NUT.
- INSTALL ONE PACKING PAD ONLY. IF GAP IS NOT REMOVED BY USING PAD, COMPRESSION BLOCKS MUST BE REPLACED.

3) With wheel clear of ground, tighten preload nut until end of preload bolt (C6SC1025–5) is level with top of nut. Lock by installing cotter pin through hole in preload nut and slot in preload bolt.

Note After a period of operation permanent set of shock absorbers can occur. If, after tightening preload nut as in Para 2.B.3), a gap is found between the compression blocks and separator plates, the shock absorber must be dismantled and the compression blocks checked for cracks or pieces missing; replace if found defective. If shock absorbers have permanent set but are otherwise serviceable, a packing pad Part No. C6UM1219–27 (see Figure 3) must be installed between the bottom surface of the lower compression block, and top surface of lower platen. Simultaneously, the pan headed screws (MS27039-5–27), and lower platen spigots Part No. C6UM1142–27 must be replaced with longer spigots Part No. C6UM1142–29 (see Figure 3) and longer pan headed screws (MS27039-5–31), using existing nuts. After reinstalling shock absorber, adjust preload nut as in Para 2.B.3).

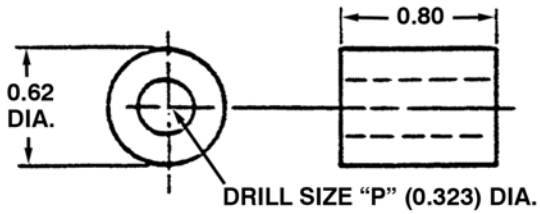
4) Lower aircraft to ground. Install main gear/fuselage fairings.

Note For aircraft incorporating Mod 6/2340, the fairings have rubber washers installed, torque the screws to compress the rubber washers (which are underneath the metal washers) to approximately one-half of their original thickness. For fairings without rubber washers (Pre Mod 6/2340) torque the #10 screws to 10 – 16 in–lbs.



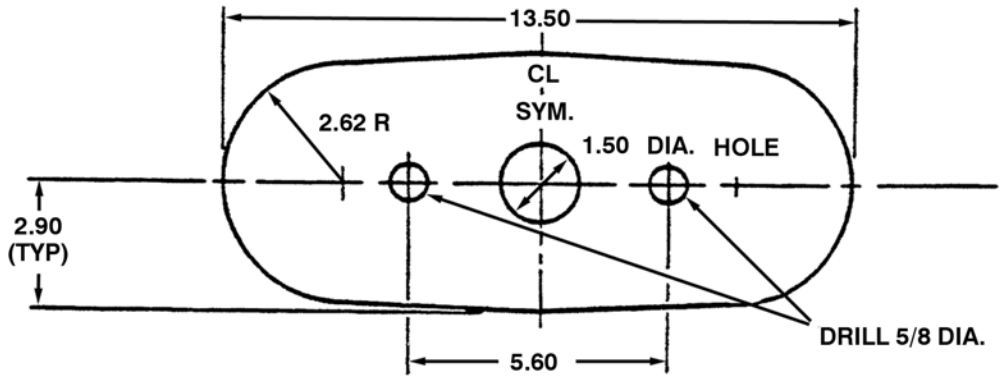
Shock Absorber Installation
Figure 2

SPIGOT (C6UM1142–29)



MATERIAL: ALUMINUM ALLOY BAR 2024–T4
SPEC: QQA–200/3
FINISH: ANODIZED (MIL–A–8625)

PACKING PAD (C6UM1219–27)



MATERIAL: 1/4 INCH THICK PHENOLIC SHEET
SPEC: MIL–P–15035 F.B.M.

Spigot and Packing Pad
Figure 3

C. Remove Upper Platen and Link Assemblies

Refer to Figure 4 for details.

1) Remove main gear/fuselage fairings.

2) Jack aircraft until wheel is almost clear of ground (refer to 07-00-00, Lifting and Shoring – General Data) and remove shock absorber (refer to Para 2.A).

3) Support aircraft on suitable cradle and/or trestle then remove jack.

4) Remove nuts and washers from upper platen mounting bolts.

5) Remove mounting bolts, washers and sleeves, then remove platen.

6) Remove nuts and washers from forward link assembly mounting bolts.

7) Remove mounting bolts, washers and sleeves, then remove forward link assembly.

8) Remove nuts and washers from aft link assembly mounting bolts.

9) Remove mounting bolts, washers and sleeves, then remove aft link assembly and shim (if shim installed).

Note Ensure shim is retained.

D. Install Upper Platen and Link Assemblies

Refer to Figure 4 for details.

1) Lightly smear platen and link assembly mounting bolts and sleeves with anticorrosion compound, Specification MIL–C–16173.

2) Install forward link assembly on landing gear forward attachment frame using sleeves, bolts, washers and nuts. Tighten nuts.

3) Temporarily install aft link assembly on landing gear aft attachment frame using sleeves.

Note Do not install shim or mounting bolts at this time.

4) Temporarily install upper platen on link assemblies using sleeves.

5) Using feeler gauge, measure and note gap between rear face of landing gear aft attachment frame and rear inner face of aft link.

6) Remove upper platen and aft link assembly.

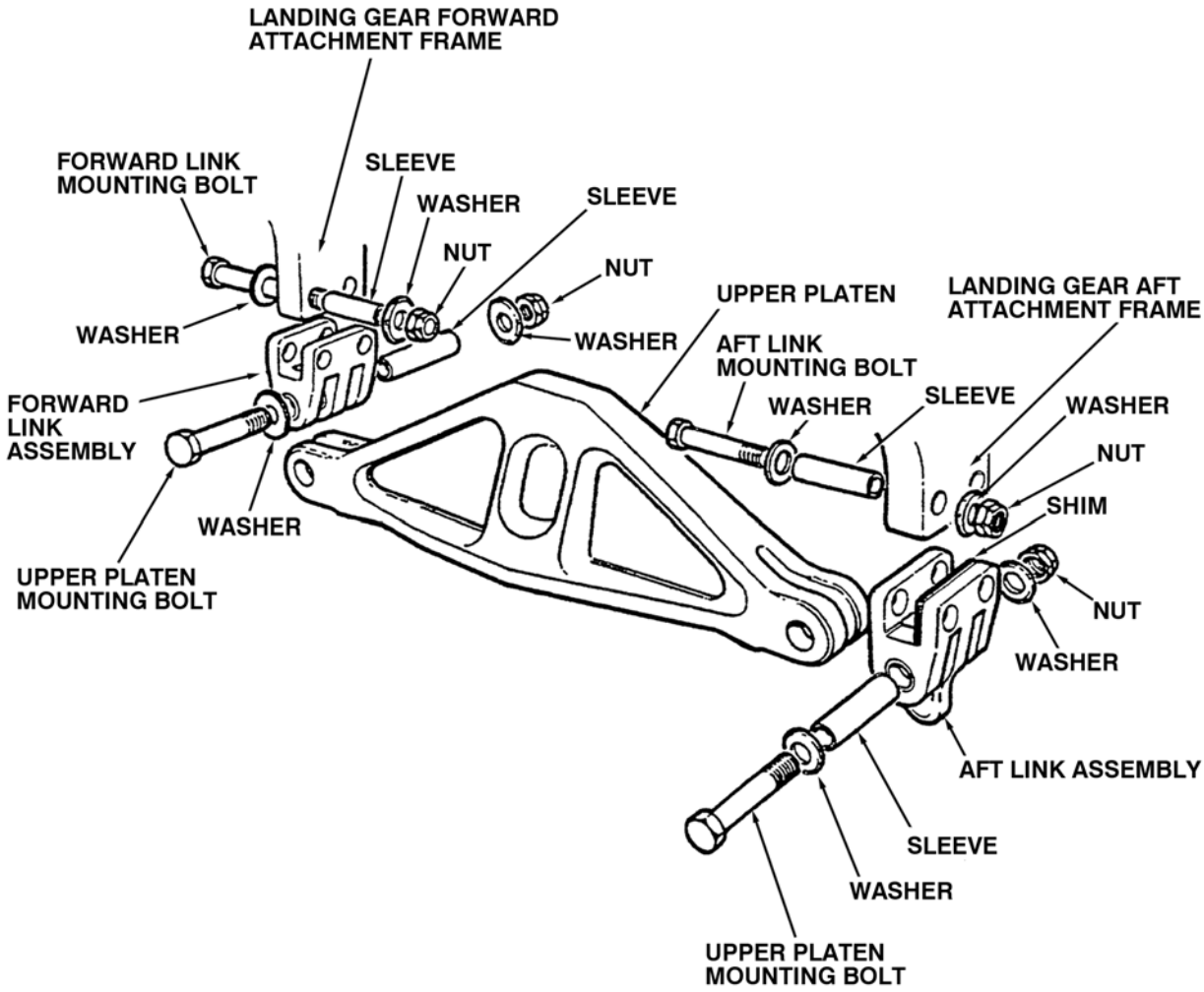
7) Prepare shim sufficient to reduce gap noted in Para 2.D.5) to 0.005 inch maximum.

8) Install aft link assembly and shim on aft landing gear attachment frame using sleeves, bolts, washers and nuts. Tighten nuts.

9) Install upper platen using sleeves, bolts, washers and nuts. Tighten nuts.

10) Jack aircraft (refer to 07-00-00, Lifting and Shoring – General Data) and remove cradle and/or trestle.

11) Install shock absorber (refer to Para 2.B).



Upper Platen Link Assemblies – Removal/Installation
Figure 4

3. Inspection/Check

A. Check Shock Absorber Blocks

1) Clean blocks of oil and grease.

2) Check compression and rebound blocks for displacement and deterioration, such as cracks, splits, or missing pieces. Pay particular attention to surface adjacent to bolt holes.

Note Some small bubbles or small clusters of bubbles may be present in the blocks due to the manufacturing processes used. These small bubbles or small clusters of bubbles are not defects, and are not cause for rejection.

3) Check compression blocks for gaps and permanent set. Check for displacement beyond edge of scuff plate (refer to Para 2.B).

Note Displacement within boundaries of scuff plate is acceptable. Discard blocks that have permanent set due to displacement beyond edge of scuff plate.

4) Check rebound blocks and plates for condition, separation and corrosion.